

ORIGINAL ARTICLE

Prospect-Theoretic Risk Assessment of Emergency Response Performance Under Procedural Dependency and Organizational Collaboration: A Dynamic Metanetwork-Based Framework

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Received: 20 January 2025 | **Revised:** 27 January 2026 | **Accepted:** 19 March 2026

Keywords: bounded rationality | dynamic metanetwork modeling and analysis | emergency response performance assessment | prospect theory | risk analysis

ABSTRACT

Emergency response performance assessment (ERPA) is critical for improving emergency management by supporting proactive risk identification and post event improvement. Existing ERPA approaches based on conceptual frameworks or system effectiveness seldom capture operational mechanisms, the decision-makers' bounded rationality, and the joint effects of procedure dependency and organizational collaboration. To address these limitations, this study develops a dynamic metanetwork-based risk analysis framework grounded in prospect theory that emphasizes multisubjective interactions within emergency response systems and distorted performance perceptions of decision-makers. The emergency response system is represented as a dynamic metanetwork with four node types, including procedure, organization, resource, and information, connected through seven subnetworks that describe system operations. Satisfaction indicators are defined at the single-procedure level and transformed using S-shaped timeliness functions to reflect bounded rationality under prospect theory, then extended to multi-procedure settings to capture procedural dependency. Key forms of organizational collaboration are identified, including procedure hand-offs, co-assignments, and resource negotiations, and collaboration indicators are constructed to measure alignment between organizational networks and these collaboration forms. Procedure timeliness and organizational collaboration are finally integrated through a fuzzy affiliation function to account for external uncertainty. The proposed framework is illustrated through a risk ERPA of the Manchester Arena attack, with sensitivity and comparative analysis demonstrating its effectiveness in complex settings.

1 | Introduction

In recent years, the frequency and complexity of emergencies have increased, posing significant challenges to public safety and social stability (R. Wang, Li, et al. 2025). Effective emergency response depends on establishing a reliable mechanism, with

emergency response performance assessment (ERPA) being a crucial element (R. Wang et al. 2022; Miao and Davlasheridze 2025). ERPA is a risk analysis technique that enables decision-makers (DMs) to identify risks in both preparedness and response process. From system perspectives, emergency response is a task-oriented, complex system where interrelated emergency response

procedures form the operational framework (Meng et al. 2024). The completion of procedures requires cross-organizational collaboration, resource allocation, and information sharing, forming a multi-subjective and multidimensional network (C. Gao et al. 2024). Horizontally, organizations spontaneously form different types of collaborative relationships, such as procedure co-assignment, procedure hand-off, or resource negotiation, which differ from typical organizational structures (Tao and Zhang 2023). Vertically, each procedure consists of actions with inherent logical relationships that make the risk disruptions in one can affect the initiation and execution of others (Cai and Golay 2023). Discrepancies in procedural timing can lead to inefficiencies, where even minor disruptions may cause significant consequences, thereby complicating ERPA under the high-risk environments. Additionally, given the complexity, urgency, and high-stakes nature of emergency situations, DMs operate under bounded rationality, making judgments with limited information, cognitive capacity, and personal preferences (Cui, Wang, et al. 2025; H. Mao and Wang 2026). These constraints, compounded by psychological and behavioral biases, lead to systematic distortions in performance perception, as explained by various behavioral decision theories, with prospect theory (Tversky and Kahneman 1974, 1992) being a representative example. In high-risk environments, DMs may assign disproportionate emphasis to certain performance ranges, amplifying perceived gains above a critical benchmark while downplaying improvements below it, resulting in nonlinear, S-shaped distortions of actual performance (R. Wang 2025).

Current methods for ERPA are primarily categorized into two groups: conceptual framework-based assessment and system effectiveness-based assessment. The conceptual framework-based assessment is the predominant subjective quantitative approach based on a conceptual indicator system. These studies focus on macro indicators such as disaster reduction (Shi et al. 2019; Qie and Rong 2022; Q. Mao et al. 2023), response quality (Wu et al. 2021; S. Wang and Du 2024; R. Wang, Shen, et al. 2025), governance level (Guo and Kapucu 2015; Zhu et al. 2023), response cost (Xu et al. 2023), and socio-economic impacts (M. Li and Cao 2019; R. Wang 2024a), often relying on expert opinions for evaluations. Additionally, bounded rationality, represented by prospect theory or cumulative prospect theory, is incorporated to address irrational decision-making in emergencies, utilizing S-shaped utility functions to reflect DMs' distort performance perceptions (Taheri et al. 2023; Q. Mao et al. 2023; D. Gao et al. 2023). However, these methods fail to analyze the underlying mechanisms of the emergency response system, with an unclear relationship between the indicator system and micro-elements (e.g., procedure, organization, resource, and information), limiting the identification of the impact of each indicator on overall performance and the development of effective optimization strategies. In contrast, system effectiveness-based assessment is an objective quantitative method that uses measurable system effectiveness indicators. These studies primarily focus on microlevel performance assessment, emphasizing failure, quality, time, and cost, and employs methods such as multi-attribute assessment (Deelstra and Bristow 2023; Chi et al. 2017), objective optimization (Hosseini et al. 2023; Sarma et al. 2022), and simulation modeling (Liu et al. 2024). However, these studies overlook the interrelationships between various performance measures and their cumulative effects, and fail to account for the

influence of psychological factors and DMs' irrational behaviors on performance distortion.

To address the above challenges and limitations, this study extends traditional metanetwork to a dynamic metanetwork (DMN) that incorporates prospect theory for ERPA considering both procedural dependency and organizational collaboration. The proposed framework models multi-subjective, multidimensional interactions in emergency response systems, encompassing procedure dependency, organization collaboration, resource allocation, and information sharing. It analyzes the gaps between actual and expected inputs of single procedure via satisfaction indicators, which are then transformed into timeliness functions that reflect the risk-induced performance distortions under prospect theory represented by S-shaped nonlinear utility. These functions are extended to capture the dependencies among multiple procedures and are complemented by a collaboration indicator quantifying the structural similarity between the organization structure defined in the official documents and the identified collaboration structure. Finally, a fuzzy affiliation function synthesizes timeliness and collaboration indicators into qualitative assessment outcomes. The main contribution of this study is summarized as follows:

- **Methodological Contributions:** This study advances ERPA research by bridging the gap between the macro-level performance indicators and the microlevel elements that generate them. Unlike conventional conceptual framework-based approaches that remain at an aggregated indicator level, the proposed framework establishes explicit, model-based linkages between procedures, organizations, resources, and information, enabling causal tracing from element-level dynamics to overall performance outcomes. Compared with system effectiveness-based approaches, the proposed framework moves beyond isolated microlevel measures by modeling their dependencies and cumulative effects, while also embedding performance distortions in prospect theory to account for DMs' bounded rationality. This integration of structural system modeling with behavioral considerations offers a mechanism-oriented basis for both diagnosing performance issues and designing targeted improvement strategies.
- **Practical Contributions:** The proposed framework is an integrated system modeling and risk analysis technique. It aligns with the operational mechanisms of typical emergency response practices and allows for expanded modeling based on actual system elements, enabling proactive diagnostics during emergency drills and retrospective investigations to pinpoint critical procedural failures. This structured and modular design facilitates systematic evaluation and supports more informed, evidence-based decision-making in emergency management.

The remaining parts of this study are organized as follows: Section 2 reviews the related literature. Section 3 introduces the proposed framework for ERPA. Section 4 demonstrates and validates the proposed framework through a scenario experiment of ERPA for the Manchester Arena attack with sensitivity and comparison analysis. Section 5 provides conclusions and future research directions.

2 | Literature Review

2.1 | Emergency Response Performance Assessment

Current methods for ERPA are primarily divided into two groups: conceptual framework-based assessment and system effectiveness-based assessment. The conceptual framework-based assessment is the mainstream subjective quantitative method, which evaluates performance based on a conceptual indicator system. In contrast, system effectiveness-based assessment is an objective quantitative method using measurable system effectiveness indicators to assess emergency response performance. Notably, in addition to pre-disaster drills and post-disaster assessment, assessing emergency plans in emergency decision-making is also a process for assessing potential emergency response performance.

Regarding the conceptual framework-based assessment, some studies establish indicator systems based on factors such as disaster reduction (e.g., casualties and environmental damage [Shi et al. 2019; Qie and Rong 2022; Q. Mao et al. 2023]), response quality (e.g., timeliness, effectiveness, rationality, and resource availability [Wu et al. 2021; S. Wang and Du 2024; R. Wang, Shen, et al. 2025]), governance level (e.g., organizational structure, responsibilities, collaboration, and information sharing [Guo and Kapucu 2015; Zhu et al. 2023]), response cost (e.g., material and personnel costs [Xu et al. 2023]), and socio-economic impact (e.g., public order and public opinion [M. Li and Cao 2019; R. Wang 2024b]). Due to the macro nature of the indicators chosen in conceptual framework-based assessments, most studies rely on expert subjective opinions for ERPA. For instance, S. Wang and Du (2024) comprehensively considered 31 indicators across 4 key aspects of oil spill emergency response, governance, support, and recovery and used a knowledge granularity model to adjust and aggregate expert opinions for ERPA. Additionally, some research develops maturity models to assess the gap between current emergency response levels and the ideal response requirements under different maturity levels (Y. Wang et al. 2023; Resende et al. 2023). For example, Y. Wang et al. (2023) proposed an emergency capability maturity model with five levels (initial, growth, standard, integration, and optimization) and developed a corresponding maturity indicator system, utilizing the AHP-cloud model to assess emergency response maturity. Notably, considering the individual psychology and DMs' irrational behavior in emergencies, some studies incorporate bounded rationality, represented by prospect theory or cumulative prospect theory, to improve assessment methods (Taheri et al. 2023; Q. Mao et al. 2023; D. Gao et al. 2023). These studies use S-shaped utility functions in (cumulative) prospect theory to distort indicator data, reflecting DMs' tendency to overestimate large-scale outcomes while underestimating smaller ones (L. Wang et al. 2015; Taheri et al. 2023). For example, Zhang et al. (2018) proposed a new ERPA method based on prospect theory, which incorporates both DMs' psychological behavior and various emergency scenarios in the decision-making process. Based on prospect theory, they developed a linear programming model to maximize the overall prospect value, which is applied to ERPA for the Barrier Lake incident and a petrochemical plant fire. While conceptual framework-based assessments provide a macro-level evaluation of emergency response performance, they fail to address the

underlying mechanisms of the response system. The link between the assessment indicators and microelements (e.g., procedure, organization, resource, and information) is unclear, hindering the ability to identify the specific impact of each indicator on overall performance and to propose effective optimization strategies.

Research on system effectiveness-based assessment primarily focuses on ERPA at the microlevel, addressing four key aspects: failure, quality, time, and cost. Methods employed include multi-attribute assessment, goal optimization, and simulation modeling. For example, Deelstra and Bristow (2023) proposed a modeling method and assessment tool to improve emergency response and preparedness. This approach considers infrastructure dependencies, repair times, and resource demands for ERPA of lifeline systems. Chi et al. (2017) introduced a timeliness function integrating time and resource satisfaction for ERPA. Hosseini et al. (2023) proposed a performance assessment and optimization method for road emergency response, accounting for uncertainties in earthquake magnitude, fault rupture location, and seismic intensity while considering recovery time and service satisfaction. Additionally, Sarma et al. (2022) introduced a hybrid multi-attribute assessment and goal optimization approach for assessing and optimizing emergency logistics. In the first stage, data collected through crowdsourcing are used to evaluate the supply priorities in affected areas based on dimensions such as casualties, medical resource shortages, public psychological stress, and residential losses. In the second stage, the priorities are integrated into a three-tier emergency solid transport model to minimize costs and meet resource demands. Liu et al. (2024) proposed a hybrid simulation and optimization approach for evaluating and optimizing emergency plans. They developed a hybrid method combining dynamic Bayesian networks and graphical evaluation and review techniques to simulate risk impact scenarios. On this basis, a multi-objective optimization model was constructed, considering resource investment, duration, and costs to optimize emergency response. It should be clarified that in system performance-based assessment, the objectives and constraints typically addressed in optimization models can, in a certain sense, be regarded as indicators. Objectives generally represent the desired optimal attributes of the system performance, while constraints are mandatory performance conditions that must be satisfied. While system effectiveness-based assessments evaluate the operational effectiveness of the emergency response system at a microlevel, they focus primarily on isolated performance measures, neglecting the interrelationships among various performance metrics and their cumulative effects. Additionally, they do not consider the influence of psychological factors and DMs' irrational behaviors on the nonlinear distortion of the utility function.

2.2 | Metanetwork Modeling and Analysis

Metanetwork modeling and analysis (MNM&A) presents a promising approach to characterizing and analyzing the complex interactions among entities in emergency response systems. Proposed by Carley and Reminga (2004), it is an emerging network analysis method designed to address complex systems involving large-scale, multidimensional, and multi-stakeholder interactions. Unlike traditional social network analysis, which focuses on identifying network members, metanetworks examine

the collaboration process, specifying which agent in which organization is assigned to a particular task, along with the resources and knowledge involved. Metanetworks provide managers with a modeling philosophy; as Pestov (2009) argue, metanetworks offer a modeling framework applicable to any social, organizational, or realistic system. Metanetworks can extract node- and network-level metrics to quantitatively assess organizational or system performance by analyzing the gaps between the resources or knowledge needed to accomplish a task and those provided (L. Li et al. 2022). Recently, MNM&A have gained attention and are applied in areas such as collaboration network risk identification (T. Wang et al. 2021; Xue et al. 2024), megaproject social responsibility analysis (Xie et al. 2022), infrastructure enhancement (Chen et al. 2025), social media opinion diffusion (Zhao et al. 2024), and organizational resilience assessment (Gu et al. 2024; Sun et al. 2025). Notably, the study of L. Li et al. (2022) is closely related to this study. They used MNM&A to assess the resilience of emergency response networks, modeling the process as a three-stage agent-task-resource-knowledge metanetwork. To quantify resilience, they selected indicators like natural connectivity, task completion speed, and the integrative metrics of task resources, knowledge needs, and waste. Their scenario experiment on the Manchester Arena attack demonstrated the method and provided policy recommendations. While the studies of MNM&A describe multi-subjective and multidimensional interactions within emergency response systems, most remain reliant on static, isolated performance evaluations and overlook the impact of dependencies between system components on overall performance.

3 | Methodology

In this section, we propose a DMN-based framework for ERPA that comprises DMN construction, procedure timeliness measurement, and organizational collaboration measurement, as illustrated in Figure 1.

3.1 | Problem Setup

The following conditions are specified to define the key elements and operational parameters of emergency response systems.

- (1) The types of procedure, organization, resource, and information included in the emergency response and the logical relationship between them;
- (2) The required time, resource quality, and quantity of each procedure;
- (3) The actual provided time, resource quality, and quantity of each procedure.

The following assumptions are established to characterize the formation and evolution of procedure timeliness in emergency response systems under DMs' behavioral mechanisms in prospect theory.

Assumption 1. The timeliness value for resource satisfaction of procedures converges to 1 when the provided quantity and quality exceed the required threshold and 0 when these are all below

the minimum threshold necessary for effectiveness. When the provided quantity and quality are between these two thresholds, the timeliness value monotonically increases from 0 to 1 as the quality and quantity of resources provided increase. The growth rate is initially slow, then rapid, and finally slow again.

Assumption 2. The timeliness value for information satisfaction of procedures equals 1 when all the required information is provided and 0 when no information is provided. When the information falls between these extremes, the timeliness value increases monotonically from 0 to 1 as the amount of information provided increases, with a rapidly accelerating growth rate.

Assumption 3. The timeliness value for completion time satisfaction of procedures converges to 0 when the actual completion time exceeds the maximum threshold. It decreases monotonically from 1 to 0 as the actual completion time increases, with the growth rate initially slow, then rapid, and finally slow again.

Assumption 4. The completion time of a procedure is inversely related to the quantity and quality of resources provided. The information satisfaction affects the this relationship by influencing the impact of the quality and quantity of available resources.

Assumption 5. The timeliness value of any response procedure is influenced by the upstream procedure, which considers only direct dependencies while excluding indirect effects.

3.2 | DMN Development

Based on the conditions outlined in Section 3.1, we develop an emergency response DMN (ER-DMN) featuring multiple node types and heterogeneous associations to capture the complex interactions in emergency response systems. Four types of nodes are identified: procedure, organization, resource, and information. These nodes are then connected by seven links representing its operational mechanisms. The developed ER-DMN provides a structured basis for characterizing collaborative and operational processes in emergency response systems, including procedure dependency, organizational coordination, resource allocation, and information sharing. In practice, such node and link types are not hypothetical constructs but are derived from the basic elements documented in emergency response plans and post-incident investigation reports. These plans and reports routinely describe which organizations undertake which procedures, what resources are mobilized, and what information is exchanged, thereby providing a solid empirical foundation for defining the semantics, directionality, and causality of the links in ER-DMN. For instance, an emergency plan may state that a municipal emergency management agency initiates evacuation procedures upon receiving real-time hazard information from monitoring units, while fire and medical departments allocate personnel and equipment according to predefined response levels. Likewise, post-incident reports typically describe the temporal sequence of decisions, interorganizational coordination mechanisms, and information feedback processes, which can be directly mapped onto corresponding nodes and causal links in the network representation.

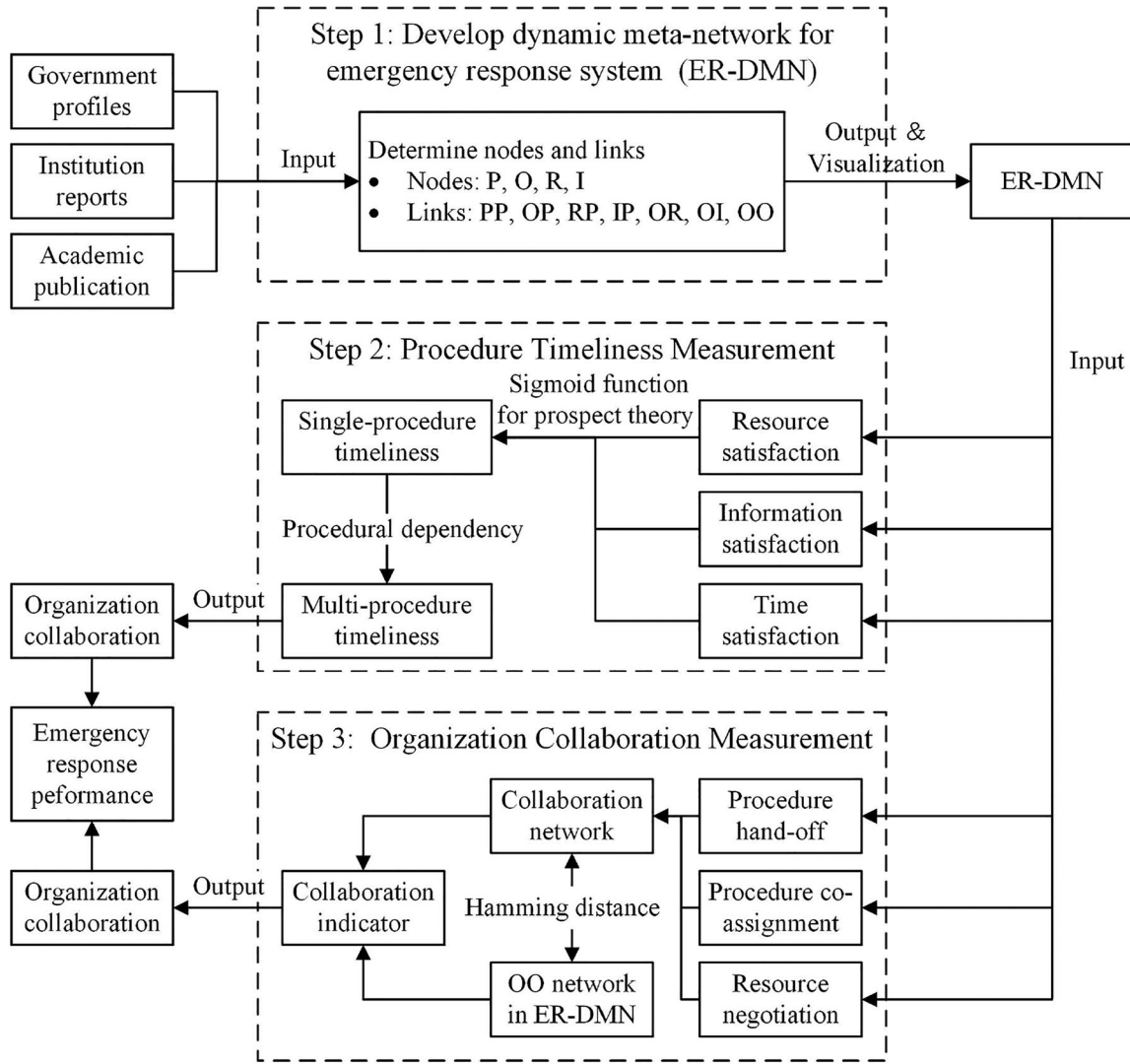


FIGURE 1 | Overview of the DMN-based ERPA framework.

We begin with defining the node sets of ER-DMN. Emergency response procedures consist of interrelated tasks derived from the emergency response, which serve as its core components (e.g., emergency assessment and rescue team assembly). Let $\mathcal{P}$ denote the set of procedure, indexed by j . Emergency organizations are structured groups of individuals, agencies, or entities responsible for preparing for, responding to, managing, and recovering from emergencies. Their functions include facilitating collaboration, allocating resources, and ensuring effective communication during crises. Let $\mathcal{O}$ denote the set of organization, indexed by i . Emergency resources refer to the tools, equipment, personnel, and supplies utilized in responding to and managing emergencies. Let $\mathcal{R}$ denote the set of resources, indexed by k . Notably, this study considers both quantitative and qualitative attributes of resources. In this case, quality is categorized into Categories I and II, with Category I being more critical than Category II. Emergency information refers to essential, timely details and updates communicated during an emergency to inform the public, operators, and authorities. This includes warnings, safety instructions, updates, and guidelines for evacuation or protective actions. Let $\mathcal{I}$ denote the set of information, indexed by l .

The seven types of associations among the defined nodes are represented as links, forming seven subnetworks in the DMN. Let PP denote the subnetwork for the dependency relationships of procedures, with its element:

$$PP_{j_1 j_2} = \begin{cases} 1, & \text{if procedure } j_1 \text{ is the upstream procedure of procedure } j_2, \\ 0, & \text{otherwise.} \end{cases}$$

This subnetwork represents the procedural workflow, where a directed link from j_1 to j_2 signifies that the completion of procedure j_1 is a prerequisite for initiating j_2 . These dependencies typically reflect temporal or logical sequencing constraints in emergency operations, such as the need for resource mobilization before on-site deployment, illustrating the causal order of tasks in emergency response.

Let OP represent the subnetwork for the participatory relationships between organizations and procedures, with its element:

$$OP_{ij} = \begin{cases} 1, & \text{if organization } i \text{ is required to participate in procedure } j, \\ 0, & \text{otherwise.} \end{cases}$$

This subnetwork connects each organization to the procedures it is responsible for or required to engage in, where a directed link from organization i to procedure j indicates that i contributes resources, personnel, or expertise to the execution of j . These links typically reflect the allocation of operational roles in emergency response.

Let RP^{QN} represent the subnetwork for the quantitative demand relationships between procedures and resources, with its element RP_{kj}^{QN} being the number of k resource required for procedure j . Let RP^{QL} denote the subnetwork for the quality demand relationships between procedures and resources, with its element RP_{kj}^{QL} being the quality of k resource required for procedure j . These subnetworks define the quantity and quality requirements of resources for emergency procedures, where links represent the required units and quality standards for each resource in a given procedure.

Let IP denote the subnetwork for the demand relationships between information and procedures, with its element:

$$IP_{lj} = \begin{cases} 1, & \text{if information } l \text{ is required for the completion of procedure } j, \\ 0, & \text{otherwise.} \end{cases}$$

This subnetwork represents the information requirements of emergency procedures, where a link from information l to procedure j indicates that l is necessary for the successful completion of j . Such informational dependencies reflect the critical role of timely and accurate data in guiding procedural execution.

Let OR^{QN} represent the subnetwork for the quantitative demand relationships between organizations and resources, with its element OR_{ik}^{QN} being the number of k resource required for organization i . Let OR^{QL} denote the subnetwork for the quality demand relationships between organizations and resources, with its element OR_{ik}^{QL} being the quality of k resource required for organization i . These subnetworks describe the resource requirements of each organization in terms of both quantity and quality.

Let OI represent the subnetwork for the awareness relationships between organizations and information, with its element:

$$OI_{il} = \begin{cases} 1, & \text{if organization } i \text{ knows information } l, \\ 0, & \text{otherwise.} \end{cases}$$

This subnetwork captures which pieces of information are known or accessible to each organization during the emergency response, where a link from organization i to information l indicates that i possesses, has been informed of, or can retrieve l when needed.

Let OO denote the subnetwork for the relationships between organizations originating from organizational structure, with its

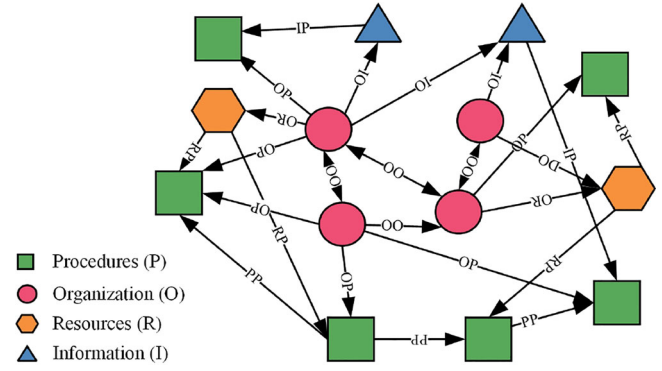


FIGURE 2 | Schematic diagram of ER-DMN.

element:

$$OO_{i_1 i_2} = \begin{cases} 1, & \text{if organization } i_1 \text{ and organization } i_2 \text{ have collaborative interaction,} \\ 0, & \text{otherwise.} \end{cases}$$

This subnetwork represents the collaborative or authoritative relationships between organizations within the emergency response system, where a link between organizations i_1 and i_2 indicates active cooperation, coordination, or command authority, reflecting formal or informal organizational structures. In Section 3.4, we further derive three types of organization collaboration based on the defined ER-DMN, including procedure co-assignment, procedure hand-off, and resource negotiation.

Figure 2 presents the conceptual diagram of the defined ER-DMN.

3.3 | Procedure Timeliness Measurement Design

In this section, we first introduce the timeliness of individual procedures. The timeliness function is formulated using satisfaction indicators and incorporates an S-shaped nonlinear utility function, which reflects DMs' psychological tendencies and potential irrational behaviors in prospect theory. We then extend this analysis to multiple dependent procedures, where the timeliness of each procedure is affected not only by factors such as resource quality and quantity, information adequacy, and completion time, but also by the timeliness of its upstream procedures. This dependency structure is explicitly considered in the aggregation of individual timeliness functions.

3.3.1 | Single Procedure Timeliness Function Construction

We first define the matrix operations that will be used later. Let $A \times B$ denote the standard matrix multiplication of matrix A with B , $A \cdot B$ is the cell by cell product of matrix A and B , and $\frac{A}{B}$ is the cell by cell division of matrix A and B . $A^\#$ denotes the binaryized version of matrix A , which is used to represent only the existence or absence of connections regardless of their magnitude. Specifically, $A^\#_{ij} = 1$ if $A_{ij} > 0$, indicating a link between nodes i and j , and $A^\#_{ij} = 0$ otherwise. This operation simplifies weighted relationships into a purely structural form, enabling analyses that focus on network topology rather than link strength. To

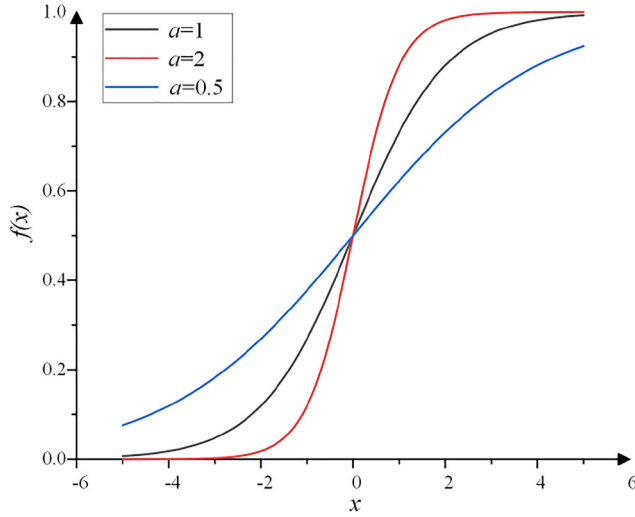


FIGURE 3 | Sigmoid function.

simplify the subsequent discussion, let q_j , Q_j , t_j , and I_j denote the provided quality and quantity of resources, actual completion time, and available information of procedure j , respectively, with corresponding notation equipped with tilde being the required, that is, $\tilde{q}_j$, $\tilde{Q}_j$, $\tilde{t}_j$, and $\tilde{I}_j$. The timeliness function of procedure j is denoted as U_j . This study utilizes the Sigmoid function to portray the impact of three factors in emergency response timeliness in terms of resource quality satisfaction (U_j^{QL}), resource quantity satisfaction (U_j^{QN}), and completion time (U_j^{CT}), except for emergency information satisfaction (U_j^I) being the quadratic form.

The Sigmoid function, as defined by Equation (1) and illustrated in Figure 3, is continuous, smooth, strictly monotonic, and centrosymmetric about (0, 0.5), with the domain of $(-\infty, +\infty)$:

$$f(x) = \frac{1}{1 + \exp(-ax)}, \quad (1)$$

where a is the inclination coefficient, the larger the value of a , the steeper the slope of the function. The choice of the Sigmoid function is motivated by both empirical observations and general bounded rationality in prospect theory, which indicate that the relationship between satisfaction indicators and emergency response performance often follows an S-shaped nonlinear pattern. This pattern reflects diminishing sensitivity at the extremes and heightened sensitivity in the mid-range: performance improvement tends to be slow when satisfaction is very low or very high, but accelerates when satisfaction is moderate. In practice, this aligns with DMs' psychological tendencies and bounded rationality under risk, where perceived utility changes gradually at the extremes and more rapidly in the intermediate range (R. Wang 2024a). Theoretically, the Sigmoid function possesses smooth, continuous, and differentiable characteristics within its domain, enabling deeper analysis of the optimal properties of the timeliness function; for example, it allows derivation of the optimal allocation of different quality resources in a single procedure. It also provides a normalized scale, enabling aggregation across multiple indicators. Notably, while this timeliness formulation is inspired by the S-shaped utility patterns also observed in prospect theory, it differs from the

empirical prospect theory value function in that it is symmetric, uncalibrated to data, and does not involve subjective probabilities. Here, the Sigmoid function is adopted as a parsimonious analytical form to represent bounded-rational utility distortions in prospect theory, rather than as a direct empirical fit.

By the assumptions in Section 3.1 and the defined ER-DMN in Section 3.2, we begin with deriving the timeliness functions for the different factors based on the corresponding satisfaction indicators.

We first present the timeliness function related to resource quality satisfaction. Resource quality satisfaction is a common indicator that measures how well the quality of allocated resources meets the specific requirements of a procedure. It reflects the extent to which the supplied resource quality supports the timely and effective completion of emergency response tasks.

Proposition 1. Let $\tilde{q}_j$ and q_j represent the required and provided resource quality for any procedure $j \in \mathcal{P}$, respectively. By Assumption 1, suppose that the minimum and maximum thresholds of procedure j for the resource quality to take effect are $\frac{1}{2}\tilde{q}_j$ and $\tilde{q}_j$, respectively. Then, the timeliness function of procedure j for resource quality satisfaction, denoted as $U_j^{QL} : \mathbb{R}^+ \mapsto [0, 1]$, in the Sigmoid form can be constructed by:

$$U_j^{QL}(RS_j^{QL}) = \frac{1}{1 + \exp\left(-\ln\left(\frac{1}{\epsilon_j^{QL}} - 1\right)^4 \left(RS_j^{QL} - \frac{3}{4}\right)\right)}, \quad (2)$$

where $RS_j^{QL} = q_j/\tilde{q}_j \geq 0$ denotes the resource quality satisfaction of procedure j and ϵ_j^{QL} is arbitrarily small. Furthermore, based on the defined ER-DMN, the resource quality satisfaction for procedure j is given by:

$$RS_j^{QL} = \sum_{k=1}^{|\mathcal{R}|} \left(\frac{((RP_{kj}^{QL})^\#)^\top \cdot (OP_{ij}^\top \times OR_{ik}^{QL})}{(RP_{kj}^{QL})^\top} \right). \quad (3)$$

Proof. The proof details are provided in the Supporting Information. $\square$

The following timeliness function quantifies resource quantity satisfaction in procedure execution. This metric assesses how well the supplied resource amounts meet the required thresholds, directly influencing the efficiency and success of emergency response operations. Accurate measurement of resource quantity satisfaction aids in identifying potential shortages or surpluses, enabling more effective resource allocation under time constraints.

Proposition 2. Let $\tilde{Q}_j$ and Q_j represent the required and provided resource quantity for any procedure $j \in \mathcal{P}$, respectively. By Assumption 1, suppose that the minimum and maximum thresholds of procedure j for the resource quantity to take effect are $\frac{1}{2}\tilde{Q}_j$ and $\tilde{Q}_j$, respectively. The timeliness function of procedure j for resource quantity satisfaction, denoted as $U_j^{QN} : \mathbb{R}^+ \mapsto [0, 1]$,

in the Sigmoid form can be expressed as:

$$U_j^{QN}(RS_j^{QN}) = \frac{1}{1 + \exp\left(-\ln\left(\frac{1}{\epsilon_j^{QN}} - 1\right) \left(RS_j^{QN} - \frac{3}{4}\right)\right)}, \quad (4)$$

where $RS_j^{QN} = Q_j/\tilde{Q}_j \geq 0$ denotes the resource quality satisfaction of procedure j and ϵ_j^{QN} is arbitrarily small. Furthermore, based on the defined ER-DMN, the resource quality satisfaction for procedure j is given by:

$$RS_j^{QN} = \sum_{k=1}^{|\mathcal{R}|} \left(\frac{((RP_{kj}^{QN})^\#)^\top \cdot (OP_{ij}^\top \times OR_{ik}^{QN})}{(RP_{kj}^{QN})^\top} \right). \quad (5)$$

Proof. The proof details are provided in the [Supporting Information](#). $\square$

We now introduce the timeliness function for information satisfaction. This indicator measures how well the provided information meets the requirements of procedures, which is essential for accurate decision-making and coordination. The quadratic form reflects the accelerating improvement in timeliness as information satisfaction increases, highlighting the increasing value of higher quality and more complete information for timely procedure completion.

Proposition 3. Let $\tilde{I}_j$ and I_j represent the required and provided information for any procedure $j \in \mathcal{P}$, respectively. By Assumption 2, the timeliness function for emergency response information satisfaction of procedure j , denoted as $U_j^I: \mathbb{R}^+ \mapsto [0, 1]$, in the quadratic form can be constructed by:

$$U_j^I(IS_j) = IS_j^2, \quad (6)$$

where $IS_j = I_j/\tilde{I}_j \geq 0$ denotes the information satisfaction of procedure j . Furthermore, based on the defined ER-DMN, the information satisfaction for procedure j is given by:

$$IS_j = \sum_{l=1}^{|\mathcal{I}|} \left(\frac{((IP_{lj})^\#)^\top \cdot (OP_{ij}^\top \times OI_{il})}{IP_{lj}^\top} \right). \quad (7)$$

Proof. The proof details are provided in the [Supporting Information](#). $\square$

We define the timeliness function for completion time satisfaction, which measures how closely the actual completion time matches the required timeframe. This metric is critical for assessing the overall efficiency and promptness of emergency response activities. Unlike other satisfaction indicators, it is influenced by resource quality, quantity, and information satisfaction.

Proposition 4. Let $\tilde{t}_j$ and t_j represent the required and actual completion time for any procedure $j \in \mathcal{P}$, respectively. By Assumption 3, the timeliness function of procedure j for completion time satisfaction, denoted as $U_j^{CT}: \mathbb{R}^+ \mapsto [0, 1]$, in the Sigmoid form

can be constructed by:

$$U_j^{CT}(TS_j) = \frac{\exp\left(-\ln\left(\frac{1}{\epsilon_j^{CT}} - 1\right) \left(TS_j - \frac{3}{4}\right)\right)}{1 + \exp\left(-\ln\left(\frac{1}{\epsilon_j^{CT}} - 1\right) \left(TS_j - \frac{3}{4}\right)\right)}, \quad (8)$$

where $TS_j = t_j/\tilde{t}_j \geq 0$ denotes the completion time satisfaction of procedure j and ϵ_j^{CT} is arbitrarily small. Furthermore, based on the defined ER-DMN and Assumption 4, the completion time satisfaction for procedure j is given by:

$$TS_j = \frac{1}{IS_j} \left(\alpha_j^1 \frac{1}{RS_j^{QL}} + \alpha_j^2 \frac{1}{RS_j^{QN}} \right) + \alpha_j^3, \quad (9)$$

where IS_j is the information satisfaction given by Equation (7), RS_j^{QL} and RS_j^{QN} are the quality and quantity satisfaction of resource given by Equations (3) and (5), respectively, and $\alpha_j^1, \alpha_j^2, \alpha_j^3 \geq 0$ are completion-time sensitivity coefficients that can be obtained from the simulation or expert judgment.

Proof. The proof details are provided in the [Supporting Information](#). $\square$

The above constructed timeliness functions are summarized in Figure 4.

This study then employs a multiplicative aggregation method to integrate the timeliness functions of the above four factors:

$$U_j(q_j, Q_j, t_j, I_j) = U_j^{QL}(RS_j^{QL})^{w_j^{QL}} \cdot U_j^{QN}(RS_j^{QN})^{w_j^{QN}} \cdot U_j^I(IS_j)^{w_j^I} \cdot U_j^{CT}(TS_j)^{w_j^{CT}}, \quad (10)$$

where $w_j^{QL}, w_j^{QN}, w_j^I$, and w_j^{CT} are weights and $w_j^{QL} + w_j^{QN} + w_j^I + w_j^{CT} = 1$. This aggregation method is motivated by the fact that a major deficiency in any single factor can drastically reduce the overall effectiveness of procedure, potentially driving the timeliness close to zero. Unlike additive models that assume factors can compensate for each other, the multiplicative approach ensures that poor performance in one aspect cannot be offset by strengths in others, thereby reflecting the dependent and non-compensatory nature of these critical components in emergency operations.

Substituting the derived timeliness function, we have:

$$U_j(RS_j^{QL}, RS_j^{QN}, IS_j, TS_j) = \left(1 + c_{j_1} \exp(a_{j_1} RS_j^{QL})\right)^{-w_j^{QL}} \cdot \left(1 + c_{j_2} \exp(a_{j_2} RS_j^{QN})\right)^{-w_j^{QN}} \cdot (IS_j^2)^{-w_j^I} \cdot \left(1 + c_{j_3} \exp(a_{j_3} TS_j)\right)^{-w_j^{CT}}, \quad (11)$$

where $a_{j_1} = -\ln\left(1/\epsilon_j^{QL} - 1\right)^4$, $a_{j_2} = -\ln\left(1/\epsilon_j^{QN} - 1\right)^4$, $a_{j_3} = -\ln\left(1/\epsilon_j^{CT} - 1\right)^{\frac{4}{3}}$; $b_{j_1} = 3/4$, $b_{j_2} = 3/4$, $b_{j_3} = 3/4$; $c_{j_1} = \exp(a_{j_1} b_{j_1})$, $c_{j_2} = \exp(a_{j_2} b_{j_2})$, $c_{j_3} = \exp(a_{j_3} b_{j_3})$.

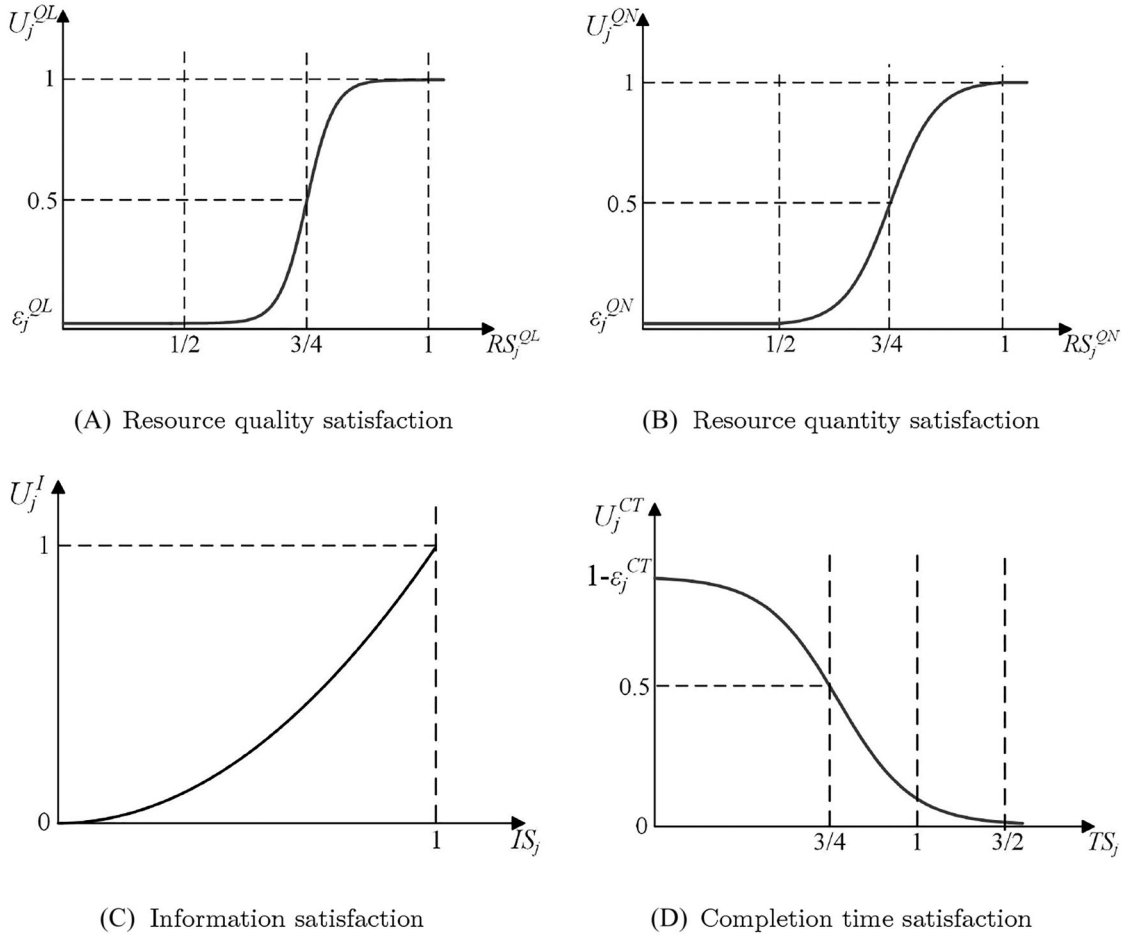


FIGURE 4 | Single procedure timeliness function considering different factors.

We further examine the optimal allocation of resource quality for single-procedure timeliness, offering insights for DMs to enhance emergency response timeliness under resource constraints. The detailed discussion is presented in the [Supporting Information](#).

3.3.2 | Multiple Procedure Timeliness Function Aggregation

In real-world emergency response scenarios, procedures are tightly dependent, forming networks where the timing and success of one task depend on others. Aggregating timeliness across multiple procedures requires capturing these dependencies to reflect system behavior and constraints. Series relationships represent sequential dependencies, where a downstream procedure cannot start or achieve timely completion without the successful and prompt execution of upstream procedures, producing cascading effects on overall timeliness. Parallel relationships capture cases in which multiple independent procedures must all be completed before subsequent steps can proceed, with overall progress limited by the slowest procedure. By using the minimum timeliness among parallel upstream procedures, the proposed framework accounts for these bottlenecks and avoids overly optimistic aggregation. Converting upstream timeliness into a compatible form enables the construction of composite timeliness functions that integrate local procedure performance with system-wide workflow constraints.

Proposition 5. Given Assumption 5 the timeliness value $U_{j'}$ of each upstream procedure of procedure j , that is, for each $j' \in \mathcal{P}'_j := \{j' : PP_{j'j} = 1\}$, the timeliness function of procedure j for series upstream procedure timeliness, denoted as $U_j^{UP} : \mathbb{R}^+ \mapsto [0, 1]$, in the Sigmoid form can be constructed by:

$$U_j^{UP}(U_{j'}^*) = \frac{1}{1 + \exp\left(-\ln\left(\frac{1}{\epsilon_{j'}^{UP}} - 1\right)\left(U_{j'}^* - \frac{1}{2}\right)\right)}, \quad (12)$$

where $\epsilon_{j'}^{UP}$ is arbitrarily small and $U_{j'}^* = \min_{j' \in \mathcal{P}'_j} \{U_{j'}\}$.

Proof. The proof details are provided in the [Supporting Information](#). $\square$

By incorporating the timeliness of upstream procedures into the timeliness function of an individual procedure, we obtain the timeliness function for procedure j considering multiple procedure aggregation, which is given by:

$$U_j(RS_j^{QL}, RS_j^{QN}, IS_j, TS_j, U_{j'}^*) = U_j^{QL}(RS_j^{QL})^{w_j^{QL}} \cdot U_j^{QN}(RS_j^{QN})^{w_j^{QN}} \cdot U_j^I(IS_j)^{w_j^I} \cdot U_j^{CT}(TS_j)^{w_j^{CT}} \cdot U_j^{UP}(U_{j'}^*)^{w_j^{UP}}, \quad (13)$$

```

1: Input: The defined ER-DMN, completion-time sensitivity coefficients  $(\alpha_j^1, \alpha_j^2, \alpha_j^3)$ , timeliness
   thresholds  $(\epsilon_j^{QL}, \epsilon_j^{QN}, \epsilon_j^{CT}, \epsilon_j^I, \epsilon_j^{UP})$ , aggregation weights  $(w_j^{QL}, w_j^{QN}, w_j^{CT}, w_j^I, w_j^{UP})$ , and upstream
   procedure sets  $\mathcal{P}'_j$  for all procedure  $j \in \mathcal{P}$ .
2: Output I: Resource quality satisfaction  $RS_j^{QL}$ , resource quantity satisfaction  $RS_j^{QN}$ , information
   satisfaction  $IS_j$ , completion time satisfaction  $TS_j$  for all procedure  $j \in \mathcal{P}$ .
3: Output II: Timeliness value for resource quality  $U_j^{QL}$ , resource quantity  $U_j^{QN}$ , information  $U_j^I$ ,
   completion time  $U_j^{CT}$ , and upstream procedure  $U_j^{UP}$  for all procedure  $j \in \mathcal{P}$ .
4: Output III: Timeliness value  $U_j$  for all procedure  $j \in \mathcal{P}$ .
5: Initialization:  $RS_j^{QL}, RS_j^{QN}, IS_j, TS_j, U_j^{QL}, U_j^{QN}, U_j^I, U_j^{CT}, U_j^{UP}, U_j \leftarrow \emptyset, \mathcal{P}^\# \leftarrow \emptyset$ .
6: while  $\mathcal{P} \neq \emptyset$  do
7:   for each  $j \in \mathcal{P}$  do
8:     if  $\mathcal{P}'_j \subseteq \mathcal{P}^\#$  then
9:       Calculate resource quality satisfaction  $RS_j^{QL}$  by Equation (3) and corresponding timeliness
       value  $U_j^{QL}$  by Equation (2) in Proposition 1.
10:      Calculate resource quantity satisfaction  $RS_j^{QN}$  by Equation (5) and corresponding timeliness
       value  $U_j^{QN}$  by Equation (4) in Proposition 2.
11:      Calculate information satisfaction  $IS_j$  by Equation (7) and corresponding timeliness value
        $U_j^I$  by Equation (6) in Proposition 3.
12:      Calculate completion time satisfaction  $TS_j$  by Equation (9) and corresponding timeliness
       value  $U_j^{CT}$  by Equation (8) in Proposition 4.
13:      if  $\mathcal{P}'_j = \emptyset$  then
14:         $U_j^{UP} \leftarrow 1$ .
15:      else
16:        Calculate timeliness value of upstream procedure  $U_j^{UP}$  by Equation (12) in Proposition 5.
17:      end if
18:      Calculate timeliness  $U_j$  by Equation (13).
19:       $\mathcal{P}^\# \leftarrow \mathcal{P}^\# \cup \{j\}$ .
20:       $\mathcal{P} \leftarrow \mathcal{P} \setminus \{j\}$ .
21:    end if
22:  end for
23: end while
24: return  $RS_j^{QL}, RS_j^{QN}, IS_j, TS_j, U_j^{QL}, U_j^{QN}, U_j^I, U_j^{CT}, U_j^{UP}, U_j$  for all procedure  $j \in \mathcal{P}$ .

```

where $w_j^{QL}, w_j^{QN}, w_j^{CT}, w_j^I, w_j^{UP}$ are weights and $w_j^{QL} + w_j^{QN} + w_j^{CT} + w_j^I + w_j^{UP} = 1$.

Algorithm 1 outlines the method for calculating procedure timeliness within the defined ER-DMN, adopting a benchmark-based approach that transforms satisfaction indicators into timeliness functions capturing DMs' nonlinear utility, and extending them from individual to dependent procedures.

3.4 | Organization Collaboration Measurement Design

In emergency situations, incident complexity, limited resources, and the need for specialized expertise often exceed the capacity

of individual agencies. Effective management requires rapid, coordinated action across organizational, departmental, and jurisdictional boundaries. Assessing interorganizational collaboration is therefore essential for evaluating emergency response and informing decision support systems. Based on the defined ER-DMN, this study identifies three types of interorganizational collaboration: procedure co-assignment, procedure hand-off, and resource negotiation, representing the primary collaboration scenarios.

Task co-assignment captures the collaboration between organizations jointly responsible for the same emergency procedure. It requires clear task allocation to ensure adequate resources and expertise while minimizing oversights or conflicts. Shared

responsibility implies the need for coordination, communication, and resource alignment to achieve effective and timely completion.

Definition 1. For any organization $i_1, i_2 \in \mathcal{O}$ (i.e., $i_1 \neq i_2$), if organization i_1 and i_2 are assigned to complete the same procedure $j \in \mathcal{P}$, then we say there is a procedure co-assignment relationship between i_1 and i_2 .

The procedure co-assignment relationship between organizations in the defined ER-DMN is given by:

$$PCA = (OP) \times (OP)^\top, \quad (14)$$

where $PCA_{i_1 i_2}$ gives the number of procedure co-assignment interactions between organizations i_1 and i_2 , particularly, the diagonal elements $PCA_{i_1 i_1}$ represent the number of procedures in which organization i_1 participates in.

Procedure hand-off represents the sequential collaboration between organizations involved in consecutive emergency procedures with upstream and downstream dependencies, transferring tasks or responsibilities from one team or department to another within or across organizations. This relationship captures the transfer of information, resources, or operational control, highlighting coordination points essential for continuity and efficiency in emergency response. It also ensures that subsequent parties are informed of the actions and decisions of their predecessors.

Definition 2. For any organization $i_1, i_2 \in \mathcal{O}$ (i.e., $i_1 \neq i_2$), if organization i_1 and i_2 are assigned to complete procedure j_1 and j_2 , respectively, where procedure j_1 and j_2 exist upstream and downstream relationships, then we say there is a procedure hand-off relationship between organization i_1 and i_2 .

The procedure hand-off relationship between organizations in the defined ER-DMN is given by:

$$PHO = (OP) \times PP \times (OP)^\top, \quad (15)$$

where $PHO_{i_1 i_2}$ gives the number of procedure hand-off interactions between organizations i_1 and i_2 , with the diagonal elements $PHO_{i_1 i_1}$ representing the number of times organization i_1 participates in both upstream and downstream procedures.

Resource negotiation captures the interaction between organizations when one assigned to a procedure lacks required resources while another possesses them. This relationship reflects the need for interorganizational coordination to reallocate or share resources, ensuring procedural requirements are met despite local shortages. It highlights key collaboration mechanisms within the emergency response network, supporting efficient resource use and overall system resilience.

Definition 3. Given organization $i_1, i_2 \in \mathcal{O}$ (i.e., $i_1 \neq i_2$), organization i_1 is assigned to complete procedure $j \in \mathcal{P}$ while i_2 is not, and there exists resource k required by procedure j that organization i_1 cannot satisfy the demand but organization i_2 have. Then, we say there is a resource negotiation relationship between organization i_1 and i_2 .

The resource negotiation relationship between organizations in the defined ER-DMN is given by:

$$RN = OP \times Z^\top \times OR^\top, \quad (16)$$

$$Z_{kj} = \begin{cases} 1, & \text{if } [OR^\top \times OP]_{kj} > 0 \wedge [RP]_{kj} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

where $RN_{i_1 i_2}$ gives the number of resource negotiation interactions between organizations i_1 and i_2 with the diagonal elements $RN_{i_1 i_1}$ representing the number of times organization i_1 uses its own resources to fulfill procedure requirements.

This study introduces an organizational collaboration indicator to assess emergency response collaboration performance, based on the structural similarity between identified collaboration structures and the OO subnetwork within the defined ER-DMN, using the Hamming distance. The collaboration indicator (EC) is calculated as follows:

$$EC = 1 - \frac{\sum_{i_1=1}^{|\mathcal{O}|} \sum_{i_2=1}^{|\mathcal{O}|} ((PCA + PHO + RN)_{i_1 i_2})^\# \cdot T_{i_1 i_2}}{|\mathcal{O}| \times (|\mathcal{O}| - 1)}, \quad (18)$$

where

$$T_{i_1 i_2} = \begin{cases} 1, & \text{if } OO_{i_1 i_2} = 0, \\ 0, & \text{otherwise,} \end{cases} \quad \text{and } U_{i_1 i_2} = \begin{cases} 1, & \text{if } S_{i_1 i_2} = 0, \\ 0, & \text{otherwise,} \end{cases}$$

A higher EC value indicates closer alignment between planned collaboration mechanisms and actual interactions, providing a measure of coordination within the emergency response system. This metric captures the structural similarity between the combined collaboration patterns, procedure co-assignment, procedure hand-off, and resource negotiation, and the actual organizational interaction network represented by the OO subnetwork. It quantifies the proportion of matching and mismatching links between the modeled collaboration structures and observed organizational relationships, yielding a normalized score that reflects the correspondence between defined collaboration patterns and real-world emergency response coordination. Algorithm 2 details the procedure for calculating organizational collaboration performance based on the defined ER-DMN.

3.5 | Emergency Response Performance Aggregation

In this section, we aggregate the previously constructed indicators of procedure timeliness and organizational collaboration into a comprehensive indicator with qualitative assessment values to address uncertainty in emergency response. Both timeliness and collaboration are benefit-oriented, with higher values indicating better performance. Therefore, we apply the multiplicative aggregation method to combine them into global effectiveness GE , as discussed previously:

$$GE = U^{w_U} \cdot EC^{w_{EC}}, \quad (19)$$

where w_U and w_{EC} are weights and $w_U + w_{EC} = 1$.

-
- 1: **Input:** The defined ER-DMN.
 - 2: **Output:** Organization collaboration performance EC .
 - 3: **Initialization:** $PCA, PHO, RN, EC \leftarrow \emptyset$.
 - 4: Identify procedure co-assignment relationship PCA by Equation (14) in Definition 1.
 - 5: Identify procedure hand-off relationship PHO by Equation (15) in Definition 2.
 - 6: Identify resource negotiation relationship RN by Equations (16) and (17) in Definition 3.
 - 7: Calculate organization collaboration performance EC by Equation (18).
 - 8: **return** EC .
-

Emergency response performance is influenced by multiple unpredictable external factors, making deterministic conclusions based solely on global effectiveness unreliable. To capture the inherent vagueness in expert judgments and linguistic performance descriptions, we employ fuzzy affiliation functions, which assign a degree of membership between 0 and 1 linking quantitative indicators to qualitative categories such as good, medium, and poor. These functions mathematically express the extent to which a numerical value corresponds to a qualitative label, even when the fit is partial. This approach interprets continuous indicator values within human-understandable semantic ranges, bridging quantitative results and qualitative risk perceptions, and enhancing the robustness of ERPA. From a qualitative perspective, the fuzzy affiliation of a good outcome $\tilde{p}_1(GE)$ increases with GE . Conversely, the fuzzy affiliation of a poor response outcome $\tilde{p}_3(GE)$ increases as GE decreases. When GE is around 0.5, the fuzzy affiliation of a medium response $\tilde{p}_2(GE)$ is highest. Thus, $\tilde{p}_1(GE)$ increases with GE , $\tilde{p}_2(GE)$ first increases and then decreases, and $\tilde{p}_3(GE)$ decreases as GE increases. These monotonic and unimodal patterns align with intuitive expert reasoning: higher effectiveness corresponds to good, lower effectiveness to poor, and intermediate effectiveness to medium. To capture these trends while maintaining computational simplicity and analytical tractability, we adopt quadratic functions rather than more complex piecewise functions, yielding the following expressions:

$$\tilde{p}_1(GE) = GE^2, \quad \tilde{p}_2(GE) = -4GE^2 + 4GE, \quad \tilde{p}_3(GE) = -GE^2 + 1. \quad (20)$$

The fuzzy affiliation function is then normalized as shown in Equation (21) with corresponding function image showing in Figure 5.

$$\begin{aligned} p_1(GE) &= \frac{\tilde{p}_1(GE)}{-4GE^2 + 4GE + 1} \\ p_2(GE) &= \frac{\tilde{p}_2(GE)}{-4GE^2 + 4GE + 1} \\ p_3(GE) &= \frac{\tilde{p}_3(GE)}{-4GE^2 + 4GE + 1} \end{aligned} \quad (21)$$

Finally, the qualitative assessment for a given GE is determined by selecting the category with the highest fuzzy affiliation value. This ensures that each quantitative effectiveness score

is consistently mapped to its most representative qualitative label, bridging numerical evaluation and human-interpretable performance categories. Notably, procedure timeliness can be assessed qualitatively using the same affiliation function in Equation (21).

3.6 | Implementation Steps

In this section, we present the implementation steps of the proposed ER-DMN-based ERPA framework and highlight key considerations for its application. Procedure 1 summarizes three main stages: constructing the ER-DMN, assessing emergency response performance, and identifying issues based on the results. Given the dynamic and unpredictable nature of emergencies, the proposed framework offers a flexible and extensible analytical tool. Effective use, however, requires a clear understanding of its components and the ability to adapt it to diverse contexts. A primary consideration is the construction of a well-structured dataset that accurately reflects the architecture and operational mechanisms of emergency response system. The dataset should allow adjustment of node and link granularity according to research objectives, extending beyond the entities defined in this study, and should be compiled from authoritative sources such as government records, professional association reports, and academic publications. A second consideration is the scope of applicability. The framework is particularly suited for targeted diagnostics, such as identifying deficiencies in emergency drills, where ER-DMN data can be collected during the exercise and used immediately afterward to construct the proposed framework, enabling clear, efficient, and cost-effective identification of shortcomings. The same approach can be applied retrospectively to specific emergency events to support investigations by pinpointing critical procedures that contributed to performance failures. In both proactive and retrospective applications, the structured and modular design of the proposed framework facilitates systematic evaluation and informed decision-making.

4 | Scenario Experiment

This section presents a scenario-based ERPA experiment using the Manchester Arena attack to demonstrate the applicability of the proposed framework. The incident, which occurred on

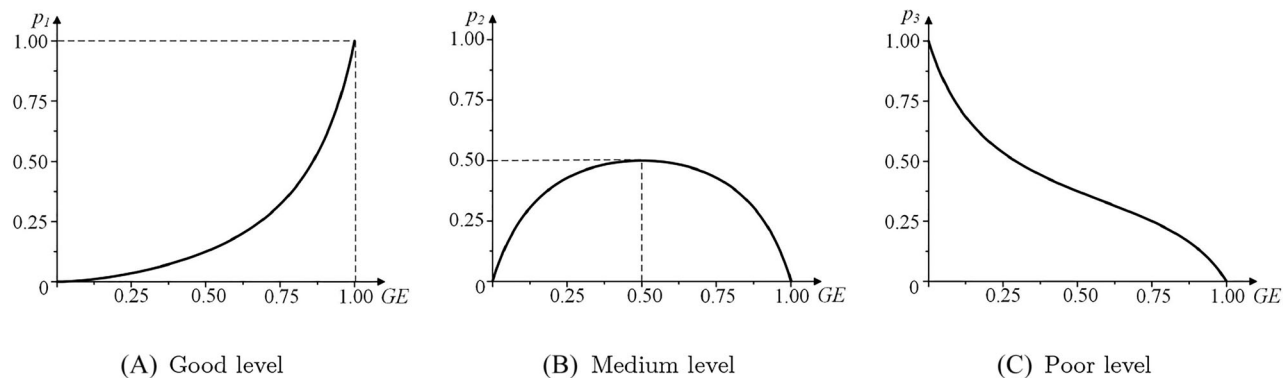


FIGURE 5 | Fuzzy affiliation function for qualitative assessment of emergency response performance.

Procedure 1 | Implementation steps of the proposed ER-DMN-based ERPA framework

- 1: **Step 1: Construct emergency response dynamic metanetwork.**
- 2: Collect data from contingency plans, official inquiry reports, and other authoritative sources.
- 3: Identify nodes of the defined node sets: procedures $\mathcal{P}$, organizations $\mathcal{O}$, resources $\mathcal{R}$, and information $\mathcal{I}$, and links of the defined link sets: PP , OP , RP , IP , OR , OI , and OO .
- 4: Determine modeling parameters: minimum and maximum thresholds of factors for procedures to take effect, completion-time sensitivity coefficients, and aggregation weights.
- 5: **Step 2: Assess emergency response performance.**
- 6: Calculate procedure timeliness based on Algorithm 1.
- 7: Calculate organization collaboration based on Algorithm 2.
- 8: Aggregate procedure timeliness and organization collaboration performance into global effectiveness according to Equation (19).
- 9: Assess qualitative semantic values for global effectiveness and procedure timeliness based on Equation (21).
- 10: Output the performance results of procedure timeliness, organization collaboration performance, global effectiveness, and their qualitative assessment semantic values.
- 11: **Step 3: Identify issues based on emergency response performance results.**
- 12: Diagnose critical bottlenecks and prioritize improvement actions via the assessment results.

May 22, 2017, resulted in 22 fatalities near the ticket office and was classified as a serious incident by British authorities. The scenario is mainly constructed based on detailed information from official inquiry reports (Saunders 2022), extracting emergency response procedures, organizations, resources, and data from these sources. Building upon this structured scenario, the ability of the proposed framework to accurately identify critical decisions and priorities, as outlined in the official 5-year government investigation findings, is assessed. Additionally, this section includes a sensitivity analysis and a comparative analysis with a baseline model to further support of the credibility of the proposed framework.

4.1 | Data Collection and DMN Extraction

The first step in implementing the proposed framework is the construction of ER-DMN, which requires the systematic extraction of detailed data on all relevant nodes and their relationships. In this study, the identification of emergency response procedures, organizations, resources, information, and their associated relationships is carried out through a comprehensive manual review of authoritative government inquiry reports (Saunders

2022), rather than relying on assumptions or secondary summaries. This ensures that the network structure is grounded in factual and verifiable evidence from the actual event. Through this process, a total of 22 emergency response procedures, 7 organizations, 15 resources, and 12 pieces of information are identified, as summarized in the [Supporting Information](#). Based on these identified nodes, seven distinct types of associations are established, resulting in a total of 424 links. These nodes and links are subsequently imported into the ORA software environment to construct ER-DMN for the Manchester Arena attack. The resulting model comprises seven sub-networks, as illustrated in Figure 6, while the temporal and logical sequence of the response procedures is represented separately in the response procedure flow diagram shown in Figure 7.

Because the government inquiry report does not provide detailed, quantitative information regarding the specific demand and supply of each type of resource, this study supplements the missing data by consulting additional reports. These materials form the basis for estimating both the demand and supply of critical resources during the emergency response, as summarized in the [Supporting Information](#). Notably, by defini-

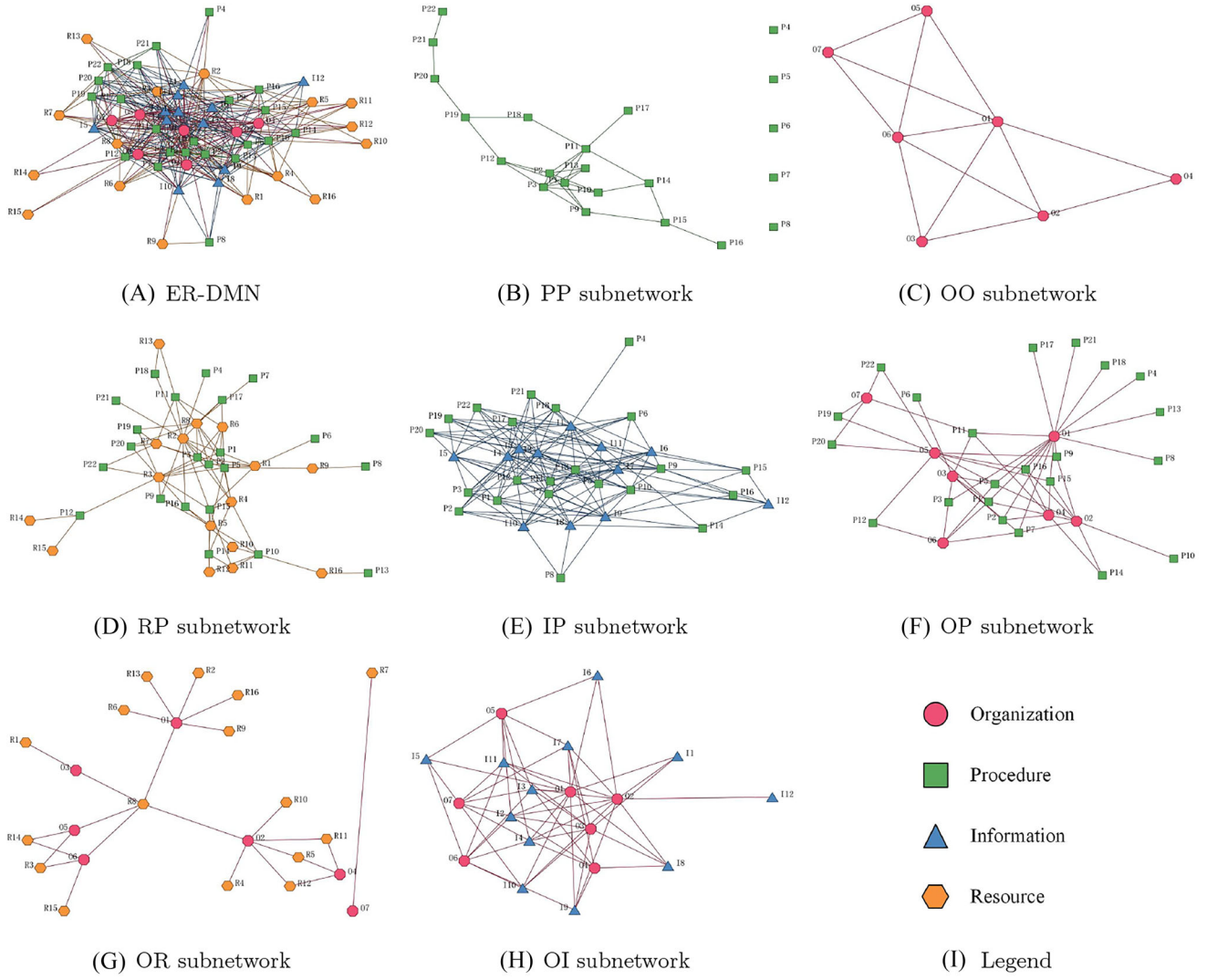


FIGURE 6 | ER-DMN of the Manchester Arena attack.

tion, the required quality of a given resource can be directly derived from its required quantity. Furthermore, to configure the proposed framework with realistic and context-appropriate settings, the necessary model parameters are based on Chi et al. (2017) combined with the judgments of experienced emergency management experts. Specifically, the weights assigned to the timeliness-related factors ($w_j^{QL}, w_j^{QN}, w_j^{CT}, w_j^I, w_j^{UP}$) are set to be equal, with each assigned a value of 0.2. Similarly, the weights for timeliness (w^U) and collaboration performance (w^{EC}) in the overall effectiveness aggregation are also set equally at 0.5. The minimum thresholds for all timeliness factors ($\epsilon_j^{QL}, \epsilon_j^{QN}, \epsilon_j^{CT}, \epsilon_j^I, \epsilon_j^{UP}$) are uniformly set to 0.001. The completion-time sensitivity coefficients are specified as $\alpha_j^1 = 0.25$, $\alpha_j^2 = 0.5$, and $\alpha_j^3 = 0$.

4.2 | Result Analysis

The satisfaction indicators for the quality and quantity of resources, information availability, and completion time for all procedures are calculated and presented in Table 1. Based on

these indicators, the timeliness of each procedure and their corresponding fuzzy affiliation values are derived, as shown in Table 2. The results reveal that procedures P5 and P9 exhibit poor timeliness, while P10, P12, P14, P15, and P16 fall within the medium timeliness range.

A closer examination indicates that the poor timeliness of P5 primarily stems from the lack of required resources and ineffective, delayed communication among executives of various emergency response organizations during the early stage of the incident. The medium timeliness of P10 is largely due to the insufficient number of police officers and the delayed arrival of specialized units. For P12, the failure of North West Fire Control and Greater Manchester Fire and Rescue Service to promptly communicate the fire staging location delayed the deployment of fire-related resources, resulting in inadequate resource availability. The poor timeliness of P9, P14, and P15 is mainly attributable to shortages of essential medical resources, such as first aid kits, stretchers, and ambulances, which also adversely impacted the timeliness of P16. Although P19, P20, and P22 achieved good timeliness ratings, they were nonetheless constrained by the absence of relevant emergency information. The

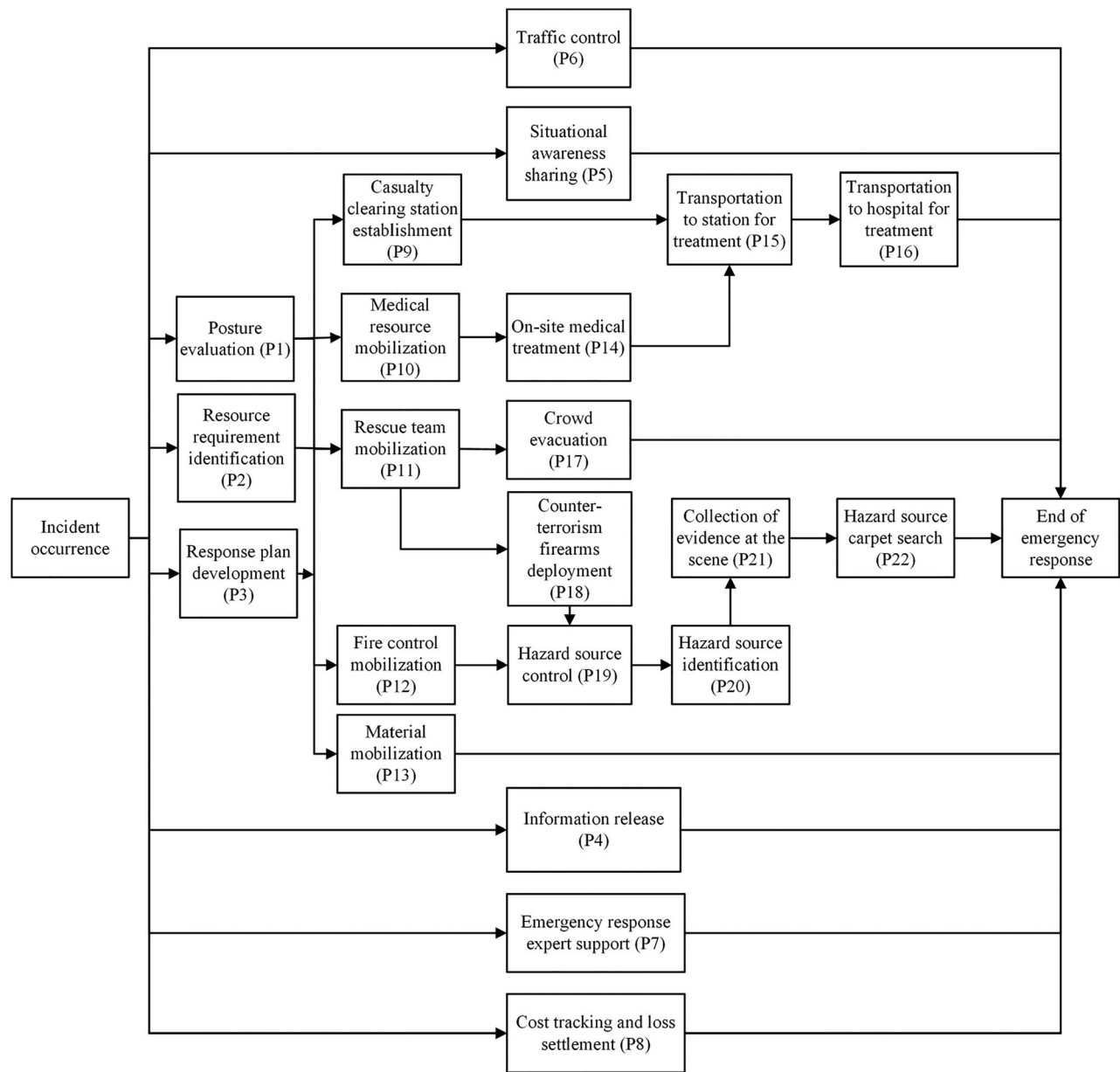


FIGURE 7 | Emergency response procedural flow of the Manchester Arena attack.

calculated organizational collaboration score for the Manchester Arena attack is 0.673, indicating a medium level of collaboration performance. Overall, the emergency response performance is assessed as medium, with notable deficiencies in situational awareness sharing, interorganizational information exchange, and the availability of emergency medical resources.

These findings are consistent with the conclusions of the official government inquiry into the attack, indicating that the proposed framework can effectively identify critical decisions and priorities comparable to those revealed through a multiyear investigation. This alignment underscores the reliability of the proposed framework when applied to models grounded in actual incident conditions. Moreover, the proposed framework offers a highly efficient diagnostic capability: once implemented, it can pinpoint issues and guide further analysis within days, in sharp contrast to the extended timelines of comprehensive post-

incident reviews. This accelerated process enhances the practical value of the proposed framework, enabling timely and evidence-based decision-making in both real-world emergency responses and training exercises.

4.3 | Sensitivity Analysis

This section presents a sensitivity analysis to evaluate the robustness of the proposed framework to variations in key parameters. Specifically, we assess the impact of changes in the thresholds for satisfaction indicators and timeliness limits. These parameters are chosen due to their significant role in shaping the timeliness function slope, which is critical for capturing distorted performance perceptions of DMs. The analysis is conducted by adjusting these parameters within a reasonable range and observing their effects on overall performance (Cui, Zhu, et al. 2025).

TABLE 1 | Results of satisfaction indicators of procedures in the Manchester Arena attack.

Procedure	Satisfaction			
	Resource quality	Resource quantity	Information	Completion time
P1	1.000	1.000	1.000	1.000
P2	1.000	1.000	1.000	1.000
P3	1.000	1.000	1.000	1.000
P4	1.000	1.000	1.000	1.000
P5	1.203	0.500	1.000	1.208
P6	1.000	0.880	0.833	1.141
P7	1.000	1.500	1.000	0.917
P8	1.000	1.000	1.000	1.000
P9	1.000	0.500	1.000	1.250
P10	1.026	0.700	1.000	1.101
P11	1.382	0.844	1.000	0.977
P12	0.783	0.938	0.800	1.232
P13	1.000	0.833	1.000	1.050
P14	1.030	0.625	1.000	1.143
P15	1.074	0.667	1.000	1.108
P16	0.907	0.800	1.000	1.088
P17	1.000	1.000	1.000	1.000
P18	1.000	1.000	1.000	1.000
P19	1.000	1.000	0.833	1.100
P20	1.000	1.000	0.833	1.100
P21	1.000	1.000	1.000	1.000
P22	1.000	1.000	0.833	1.100

Note: Values in bold indicate less/greater than 1 standard deviation(s) above the mean.

Figure 8 shows the sensitivity analysis of timeliness values under varying satisfaction indicator thresholds, ranging from -20% to $+20\%$ relative to the baseline. For most procedures (P1–P4, P6–P8, P11, P13, P16–P22), timeliness remains largely unchanged with threshold variations, indicating low sensitivity. However, several procedures (notably P5, P9, P12, P14, and P15) exhibit significant variations, with marked decreases in timeliness as thresholds increase. Additionally, threshold adjustments cause performance level shifts for certain procedures. For instance, a 20% reduction for P5 shifts its rating from poor to medium; for P14 and P15, a 20% increase downgrades their ratings from medium to poor; and for P16, a 10% or 20% reduction improves its rating from medium to good. These patterns reflect the relationship between the performance level of each procedure and the satisfaction threshold. Procedures near the threshold are more sensitive to slope changes; increasing the threshold steepens the slope, amplifying perceived timeliness differences, while decreasing the threshold flattens the slope, reducing these differences. This effect is particularly pronounced for poorly performing procedures, where small changes in evaluation parameters lead to disproportionately large shifts in perceived timeliness. This mechanism explains the distorted performance perceptions of DMs, where minor performance changes may appear amplified or compressed depending on the parameter settings.

Figure 9 shows the sensitivity analysis of timeliness values under varying timeliness limits, including 0.01, 0.005, 0.001, 0.0005, 0.0001. For most procedures (P1–P4, P6–P8, P10–P13, P16–P19, P21–P22), timeliness remains largely unchanged across different timeliness limits, indicating low sensitivity. However, several procedures (notably P5, P14, P15, and P20) exhibit noticeable variations. Performance level shifts occur for these procedures: for P5, increasing timeliness limit from 0.001 to 0.01 changes its rating from poor to medium; for P14 and P15, larger values also improve ratings from poor to medium; and for P20, increasing timeliness limit from the baseline to 0.01 downgrades its rating from good to medium. These patterns can be explained by the fact that larger timeliness limits flatten the slope of the timeliness (sigmoid) function, thereby reducing the rate at which timeliness changes near the inflection region. This slope flattening compresses differences in actual performance, leading to smaller perceived performance gaps among procedures. Consequently, procedures with baseline performance levels far from the inflection point or close to the upper asymptote remain largely unaffected, whereas those near the inflection zone are more sensitive to timeliness limit changes, potentially resulting in reclassification of their performance levels.

Overall, the sensitivity analysis confirms that the proposed framework is largely robust to variations in satisfaction

TABLE 2 | Results of procedure timeliness of the Manchester Arena attack.

Procedure	Timeliness					Fuzzy affiliation				
	Resource quality	Resource quantity	Completion time	Information	Upstream	Overall	Poor	Medium	Good	
	U_j^{QL}	U_j^{QN}	U_j^{CT}	U_j^I	U_j^{UP}	U_j	p_1	p_2	p_3	
P1	1.000	1.000	0.909	1.000	1.000	0.981	0.036	0.070	0.894	Good
P2	1.000	1.000	0.909	1.000	1.000	0.981	0.036	0.070	0.894	Good
P3	1.000	1.000	0.909	1.000	1.000	0.981	0.036	0.070	0.894	Good
P4	1.000	1.000	0.909	1.000	1.000	0.981	0.036	0.070	0.894	Good
P5	1.000	0.001	0.596	1.000	1.000	0.226	0.558	0.412	0.030	Poor
P6	1.000	0.973	0.732	0.694	1.000	0.868	0.169	0.314	0.517	Good
P7	1.000	1.000	0.956	1.000	1.000	0.991	0.018	0.035	0.947	Good
P8	1.000	1.000	0.909	1.000	1.000	0.981	0.036	0.070	0.894	Good
P9	1.000	0.001	0.500	1.000	0.840	0.211	0.573	0.400	0.027	Poor
P10	1.000	0.201	0.798	1.000	0.840	0.670	0.293	0.469	0.238	Medium
P11	1.000	0.931	0.925	1.000	0.840	0.937	0.098	0.190	0.711	Good
P12	0.713	0.994	0.541	0.640	0.840	0.729	0.262	0.441	0.297	Medium
P13	1.000	0.908	0.863	1.000	0.840	0.920	0.119	0.228	0.653	Good
P14	1.000	0.031	0.729	1.000	0.819	0.449	0.401	0.497	0.101	Medium
P15	1.000	0.092	0.788	1.000	0.456	0.505	0.372	0.500	0.128	Medium
P16	0.987	0.799	0.816	1.000	0.505	0.799	0.220	0.391	0.388	Medium
P17	1.000	1.000	0.909	1.000	0.819	0.942	0.092	0.179	0.729	Good
P18	1.000	1.000	0.909	1.000	0.819	0.942	0.092	0.179	0.729	Good
P19	1.000	1.000	0.799	0.694	0.822	0.854	0.181	0.333	0.487	Good
P20	1.000	1.000	0.799	0.694	0.773	0.844	0.189	0.345	0.466	Good
P21	1.000	1.000	0.909	1.000	0.766	0.930	0.107	0.207	0.686	Good
P22	1.000	1.000	0.799	0.694	0.815	0.853	0.182	0.334	0.484	Good

Note: Values in bold indicate less than 1 standard deviation(s) above the mean.

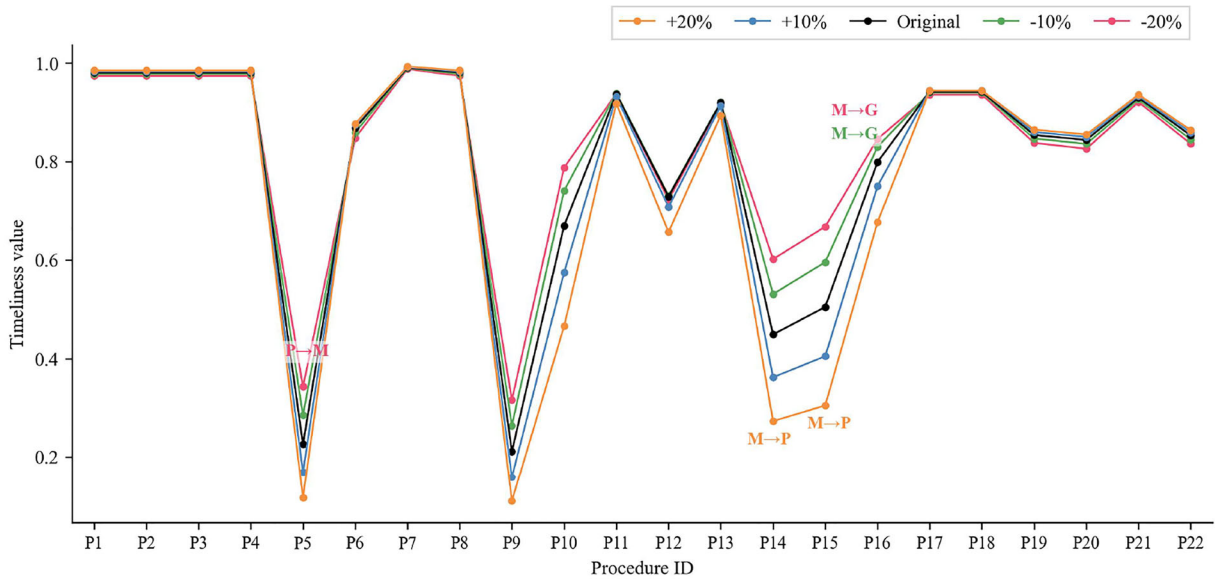


FIGURE 8 | Sensitivity results of timeliness values under varying satisfaction indicator thresholds.

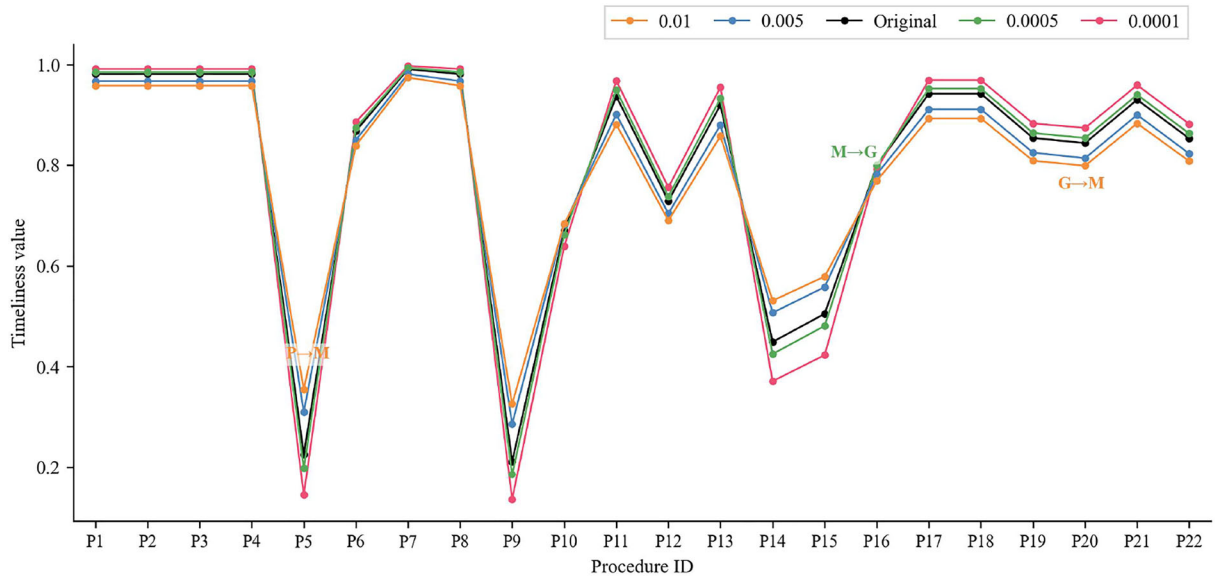


FIGURE 9 | Sensitivity results of timeliness values under varying timeliness limits.

thresholds and timeliness limits, as most procedures maintain stable timeliness values and performance levels. As expected from the structure of timeliness function, procedures with lower baseline timeliness values are more sensitive to parameter changes, while those with higher values remain largely unaffected.

4.4 | Comparison Analysis

This section compares the proposed framework with baseline models to assess its validity and effectiveness. The baseline models vary across two components: (1) timeliness function form (linear vs. Sigmoid), (2) aggregation method (additive vs. multiplicative), and (3) upstream procedure timeliness (inclusive

vs. exclusive). This comparison isolates the impact of performance distortions due to bounded rationality and procedural dependencies on ERPA outcomes.

Figure 10 compares the timeliness values of emergency response procedures obtained from different models. LN-AD-NUP represents the model with linear timeliness functions aggregated additively, excluding upstream procedural timeliness. SIG-AD-NUP refers to the model with Sigmoid-based timeliness functions aggregated additively, also excluding upstream procedural timeliness. SIG-MP-NUP is the model with Sigmoid-based timeliness functions aggregated multiplicatively, without considering upstream procedural timeliness. Finally, SIG-MP-UP is the model with Sigmoid-based timeliness functions aggregated multiplicatively, incorporating upstream procedural timeliness.

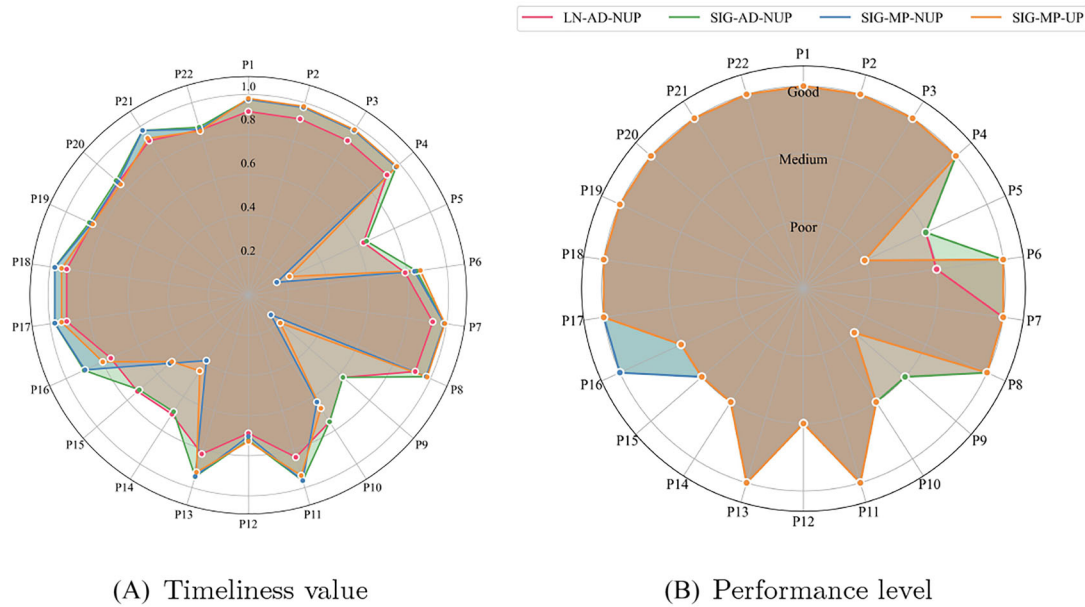


FIGURE 10 | Comparison results of timeliness values of original model and other benchmark models.

The comparison between LN-AD-NUP and SIG-AD-NUP reveals that the Sigmoid-based function generally yields higher timeliness scores, especially for procedures with low or medium baseline performance (e.g., P5, P14, P15). This results from reduced penalties in the lower range and amplified gains in the mid-range, which elevate these procedures in the performance ranking. Conversely, procedures near the upper bound (e.g., P1–P4, P17–P19) show minimal change, as their scores fall within the flat region of the Sigmoid curve, where sensitivity to slope is low. This pattern is due to the nonlinear slope of the Sigmoid function, which amplifies differences around the inflection point but compresses those near the extremes. As a result, mid-range performers experience more significant score increases, narrowing the performance gaps. This behavior mirrors DMs' perceptions, where moderate improvements in the mid-range are more noticeable, while changes at the extremes have less impact.

Switching from additive (SIG-AD-NUP) to multiplicative aggregation (SIG-MP-NUP), timeliness values for most high-performing procedures (e.g., P1–P4, P7, P8, P11, P13, P17–P19, P21–P22) remain largely unchanged, indicating their consistent strength across all factors. However, procedures with lower scores in at least one factor (e.g., P5, P9, P14, P15) experience significant declines, such as P5 dropping from 0.649 to 0.156 and P14 from 0.690 to 0.387. This occurs because of the non-compensatory nature of multiplicative aggregation, where a deficiency in one dimension disproportionately lowers the overall score, without being balanced by stronger performance in other areas. Thus, the multiplicative approach amplifies the difference between strong and weak procedures, penalizing those with bottleneck factors more heavily.

When upstream procedural timeliness is integrated into the multiplicative aggregation model (SIG-MP-UP), most high-performing procedures (e.g., P1–P4, P7, P8) show minor increases in timeliness, such as P1–P4 rising from 0.976 to 0.981, reflecting

the advantage of strong upstream performance. Notably, several procedures with previously low scores due to bottleneck factors (e.g., P5, P9, P14) also improve, for example, P5 from 0.156 to 0.226 and P14 from 0.387 to 0.449, when their upstream procedures perform more efficiently. This suggests that delays are partially mitigated by faster preceding steps. However, integrating upstream timeliness can also result in reductions (e.g., P16 drops from 0.896 to 0.799, P20 from 0.862 to 0.844) when upstream procedures are slower, propagating delays downstream. Overall, incorporating upstream timeliness more effectively captures the dependencies in emergency operations: strong upstream performance enhances downstream timeliness, while delays in earlier steps can hinder subsequent performance, even when the current step is efficient.

Overall, the proposed framework effectively captures the perceptual distortions of bounded rationality and the operational dependencies in emergency response procedures. By using a Sigmoid-based timeliness function, the model better aligns timeliness scoring with human perception, enhancing mid-range differences while compressing extremes. The multiplicative aggregation strengthens the non-compensatory nature of key factors, ensuring that deficiencies in any dimension are penalized. Incorporating upstream procedural timeliness further reflects realistic dependency chains, where delays or efficiencies in earlier steps affect downstream performance. These design choices result in a more precise, realistic, and context-sensitive evaluation of emergency response timeliness.

5 | Conclusion

This study advances ERPA research by addressing two limitations in existing approaches: the inability to jointly model the performance dependencies between emergency response procedures and the boundedly rational decision-making behavior of DMs in prospect theory under risk. To overcome these gaps, we propose a DMN-based ERPA framework that explicitly links

dependent procedural performance with organizational collaboration patterns. Methodologically, the proposed framework introduces a structured representation of emergency response operations through four node types, procedures, organizations, resources, and information, organized into seven subnetworks, thereby capturing both structural and functional aspects of the response system. Building on this representation, we define satisfaction indicators for resource quality, resource quantity, information, and completion time, and apply a Sigmoid-based transformation to reflect the nonlinear influence of these indicators on DMs' utility under bounded rationality. This yields procedure-level timeliness functions that are further extended to account for interprocedural dependencies. Meanwhile, we formalize three distinct forms of organizational collaboration, procedure co-assignment, procedure hand-off, and resource negotiation, quantified through a Hamming-distance-based indicator. By integrating timeliness and collaboration via a fuzzy affiliation function, the proposed framework enables a joint assessment of procedural timeliness and organizational coordination. The proposed framework offers a network-based approach for ERPA that integrates operational dependencies and bounded rationality in decision-making, addressing certain limitations in existing modeling efforts of emergency response systems.

The proposed framework is evaluated through a scenario-based ERPA experiment of the Manchester Arena attack, using structured data from official inquiry reports. The scenario experiment demonstrates that the proposed framework can reproduce critical decisions and priorities consistent with those identified in the official multiyear investigation, indicating its capacity to capture essential operational features of real incidents. Sensitivity analysis further examines the robustness of the proposed framework to variations in key parameters influencing the timeliness function slope, showing stable assessment trends under reasonable parameter changes. Comparative analysis against baseline models highlights the added value of incorporating bounded rationality and performance dependencies, clarifying their influence on ERPA outcomes. Together, these validation steps provide evidence of the applicability and analytical usefulness of the proposed framework in both retrospective and diagnostic contexts.

This study has several limitations that point to opportunities for future research. First, the findings are based on a specific case, highlighting the need to apply and validate the proposed framework across a wider range of emergency scenarios to assess its generalizability. Second, the proposed framework is currently tailored for offline ERPA and lacks capabilities for real-time analysis; incorporating real-time data integration would enhance its responsiveness in dynamic risk environments. Third, the timeliness function is developed through a model-based construction grounded in theoretical assumptions, which provides a family of functions with desired properties. Future work could explore deriving these timeliness functions directly from behavior experiments to better capture actual preferences, or alternatively, focus on selecting appropriate functions within the existing families based on specific decision criteria. The latter approach requiring less effort as it avoids costly experiments and concentrates on model refinement.

Author Contributions

Renlong Wang: writing – original draft, conceptualization, methodology, validation, software. **Jingyao Yue:** writing – original draft, validation, visualization. **Jintao Lu:** writing – review and editing. **Anh Ngoc Quang Huynh:** writing – review and editing. **Xueyan Shao:** validation, writing – review and editing. **Mingang Gao:** supervision, conceptualization, writing – review and editing.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Data S1